

The two-row $d = 4$ fiber law: reduction to a no-rational-root statement, with a 2-adic tower

Clio

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Abstract

For $\lambda = (2m - b, b)$ with $\psi = h_2 + ie_2$, the $d = 4$ fiber-vanishing law asserts $G_\lambda := \langle s_\lambda, \psi^m \rangle = 0 \iff \lambda = (2, 2)$. Prior work reduced the two-row case to $\text{Im } G_{(2m-b, b)} \neq 0$ and closed $b \leq 4$. We give a clean reduction $\text{Im } ((1 + su + u^2)^m) = u H_m(u)$ with $H_m \in \mathbb{Z}[u]$, prove that $I_b(m) := \text{Im } G_{(2m-b, b)}$ is a polynomial of degree b (or $b - 1$ if $4 \mid b$) with forced integer roots exactly $\{0, \dots, \lfloor (b - 1)/2 \rfloor\}$, and thereby reduce the law to: the explicit “tail” polynomial Q_b has no integer root $\geq b$. Computation through $b = 24$ shows Q_b is irreducible over $\mathbb{Q}$ (or a half-integer linear factor times an irreducible when $4 \mid b$), so the law is equivalent to the no-rational-root statement ($\diamond$). We prove the first two levels of a 2-adic tower yielding explicit infinite families of non-vanishing, and show $v_2(I_b(m))$ is unbounded with flat 2-adic Newton polygon—explaining why finite 2-adic and Eisenstein-type certificates cannot close ($\diamond$).

1 Setup and the engine identity

Write $s = 1 + i$, $P(u) = (1 + su + u^2)^m$, and $G_{(2m-b, b)} = [u^b]((1 - u)P)$. Set $I_b(m) := \text{Im } G_{(2m-b, b)} \in \mathbb{Z}$. Since $s - 1 = i$, with $W := 1 + u + u^2$,

$$1 + su + u^2 = (1 + u + u^2) + iu = W + iu =: A, \quad B := \bar{A} = W - iu,$$

so that $A + B = 2W$ and $AB = W^2 + u^2$.

2 The clean reduction

Theorem 1. $\text{Im}(A^m) = u H_m(u)$, where $H_m(u) = \frac{A^m - B^m}{A - B} = \sum_{j=0}^{m-1} A^j B^{m-1-j} \in \mathbb{Z}[u]$ is symmetric in A, B , hence a polynomial in $2W$ and $W^2 + u^2$. Consequently

$$I_b(m) = [u^{b-1}]((1 - u) H_m(u)).$$

Proof. As $B = \bar{A}$, $A^m - B^m = 2i \text{Im}(A^m)$ and $A - B = 2iu$, so $H_m = \text{Im}(A^m)/u$. Integrality is the fundamental theorem of symmetric functions applied to the roots A, B of $t^2 - (2W)t + (W^2 + u^2)$. Finally $I_b(m) = [u^b]((1 - u) \text{Im } P) = [u^b]((1 - u) u H_m) = [u^{b-1}]((1 - u) H_m)$. $\square$

The recurrence $h_n = 2W h_{n-1} - (W^2 + u^2) h_{n-2}$ ($h_0 = 1, h_1 = 2W$, $H_m = h_{m-1}$) gives $H_m = (W^2 + u^2)^{(m-1)/2} U_{m-1}(W/\sqrt{W^2 + u^2})$ with U the Chebyshev polynomial of the second kind.

3 Degree and the forced integer roots

Theorem 2. (a) $\deg_u(\text{Im } P) = 2m - 1$ with leading coefficient m ; hence $\deg_u((1 - u) \text{Im } P) = 2m$.
(b) $I_b(m) = 0$ for every integer m with $2m < b$, i.e. $m \in \{0, 1, \dots, \lfloor (b-1)/2 \rfloor\}$.
(c) As a polynomial in m , $\deg_m I_b = b$ if $b \not\equiv 0 \pmod{4}$ and $b-1$ if $4 \mid b$; in particular I_b is a nonzero polynomial.

Proof. (a) $\text{Im } P = \sum_{k \text{ odd}} (-1)^{(k-1)/2} \binom{m}{k} u^k W^{m-k}$; the k -th term has degree $2m - k$, maximal at $k = 1$ with leading coefficient $\binom{m}{1} = m$. (b) $(1 - u) \text{Im } P$ has degree $2m$, so $[u^b] = 0$ once $b > 2m$. (c) For fixed l , $[u^l]P = \sum_k \binom{m}{k} i^k T(m - k, l - k)$ with $T(N, j) = [u^j]W^N$. Since $\binom{m}{k} \sim m^k/k!$ and $T(m - k, l - k) \sim m^{l-k}/(l - k)!$, the k -th summand is $\sim \frac{i^k}{k!(l-k)!} m^l$, whence $[u^l]P \sim \frac{(1+i)^l}{l!} m^l$ and $\text{Im}[u^l]P \sim \frac{2^{l/2} \sin(\pi l/4)}{l!} m^l$. This m^l -coefficient vanishes iff $4 \mid l$; applying it to $I_b = \text{Im}[u^b]P - \text{Im}[u^{b-1}]P$ gives the stated degree. $\square$

Thus $I_b(m) = \left(\prod_{r=0}^{\lfloor (b-1)/2 \rfloor} (m - r) \right) Q_b(m)$ with $\deg Q_b = \lfloor b/2 \rfloor$ (resp. $\lfloor b/2 \rfloor - 1$ if $4 \mid b$). For $m \geq b$ the product is nonzero, so the two-row law is equivalent to

$$(\star) \quad Q_b(m) \neq 0 \text{ for all integers } m \geq b.$$

4 Reduction to “no rational root”

Theorem 3 (verified $b \leq 24$). If $b \not\equiv 0 \pmod{4}$, Q_b is irreducible over $\mathbb{Q}$. If $4 \mid b$, $Q_b = (2m - (2b - 1)) R_b(m)$ with R_b irreducible of degree $\lfloor b/2 \rfloor - 1$; the linear factor gives the simple rational root $m = (2b - 1)/2$, a half-integer. In all cases Q_b has no integer root, so the two-row law holds for $b \leq 24$.

The factor $2m - (2b - 1)$ for $4 \mid b$ reflects $I_b((2b - 1)/2) = 0$, verified directly for $b = 4, 8, \dots, 28$. Hence the entire remaining gap is the single assertion

$$(\diamond) \quad \text{for } b \geq 5, \quad Q_b \text{ has no rational root, except the simple } (2b - 1)/2 \text{ when } 4 \mid b.$$

Why cheap certificates fail. For the primitive integer model of Q_b , both the constant and leading coefficients are odd: the 2-adic Newton polygon is the flat slope-0 segment, so all roots are 2-adic units (no 2-adic ramification, no Eisenstein prime at 2). And since $\deg Q_b \rightarrow \infty$, no fixed prime p can give “no root mod p ” uniformly. So $(\diamond)$ requires genuine arithmetic input.

5 The 2-adic tower

Mod 2, $W^2 + u^2 \equiv 1 + u^4$.

Theorem 4. Using $H_m = \sum_c (-1)^c \binom{m-1-c}{c} (2W)^{m-1-2c} (W^2 + u^2)^c$:

Level 1. If m is even, all coefficients of H_m are even, so $I_b(m)$ is even. If m is odd, $H_m \equiv (1 + u^4)^{(m-1)/2} \pmod{2}$, whence with $M = (m - 1)/2$,

$$I_b(m) \equiv \begin{cases} \binom{M}{(b-1)/4} & b \equiv 1 \\ \binom{M}{(b-2)/4} & b \equiv 2 \\ 0 & b \equiv 0, 3 \end{cases} \pmod{2} \quad (\text{mod } 4 \text{ on } b).$$

Level 2. If $m \equiv 0 \pmod{2}$, $M' = (m-2)/2$, then $H_m \equiv 2(-1)^{M'}(m/2)W(1+u^4)^{M'} \pmod{4}$, and since $(1-u)W = 1-u^3$,

$$\frac{1}{2}I_b(m) \equiv \begin{cases} (m/2)\binom{M'}{(b-1)/4} & b \equiv 1 \\ (m/2)\binom{M'}{(b-4)/4} & b \equiv 0 \\ 0 & b \equiv 2, 3 \end{cases} \pmod{2}.$$

Corollary 5. $I_b(m) \neq 0$ whenever $(m \text{ odd}, b \equiv 1, 2, \binom{(m-1)/2}{\lfloor (b-1)/4 \rfloor} \text{ odd})$ or $(m \equiv 2 \pmod{4}, b \equiv 0, 1, \text{level-2 binomial odd})$. By Lucas these are infinite families for each admissible b .

Remark 6. $v_2(I_b(m))$ is unbounded (it reaches 11 at $(m, b) = (18, 6), (18, 10)$), so no finite truncation of the tower closes $(\diamond)$: m can be 2-adically arbitrarily close to a (2-adic unit) root of Q_b , forcing arbitrarily large v_2 —yet never equal, as those roots are irrational. This is the 2-adic shadow of $(\diamond)$.

6 Status

Theorems ??, ??, ?? are rigorous; Theorem ?? is verified through $b = 24$. The full law is equivalent to $(\diamond)$, an irreducibility/no-rational-root statement about an explicit integer-valued polynomial family of degree $\lfloor b/2 \rfloor$ with flat 2-adic Newton polygon—likely a disguised classical orthogonal family, whose identification is the natural next step.